

KEVIN O'DONNELL

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EDUCATION

Columbia University New York, NY
Master of Science in Biomedical Engineering, Concentration in Robotics and Control Dec 2024
Coursework: Biostatistics, Robotics Studio, Mechatronics, Tissue/Biomaterial Eng, Control Theory

University of California, Los Angeles (UCLA) Los Angeles, CA
Bachelor of Science in Honors Integrative Biology and Physiology Mar 2022
Awards: Department Honors, Alumni Scholarship National Finalist, Achievement Scholarship

PROFESSIONAL EXPERIENCE

ForUs (Generator Health) Remote
Operations Specialist Feb 2026 - Present

- Automate workflows using Python/Excel to expedite case resolution and elevate quality control.
- Develop tools to synthesize data relationships and audit across AI-driven healthcare operations.
- Train team members, coordinate daily workloads, and collaborate on SOP development.
- Resolve 100+ daily complex pharmacy cases by coordinating with insurers, providers, and patients.

Shape Memory Medical Inc. San Jose, CA
Engineering Intern Jun 2023 - Aug 2023

- Designed 3 patient-specific cardiac aneurysm CAD models in SolidWorks, 3D-printed prototypes, and integrated them into a pulsatile flow system for R&D preclinical device Verification & Validation.
- Remediated manufacturing-line stent crimper electrical circuit, enhancing operator safety & efficiency.
- Optimized laser welding using Design of Experiments, achieving a 367% increase in tensile strength.

Samuel Sia Laboratory - Microfluidics New York, NY
Graduate Research Assistant Aug 2022 - Dec 2022

- Characterized microfluidic pump designs to optimize automated wastewater SARS-CoV-2 surveillance, streamlining sample processing and accelerating early detection. Executed PCR testing.

Colwell Laboratory of Circadian and Sleep Medicine Los Angeles, CA
Undergraduate Research Assistant Jul 2019 - Jun 2022

- Led a 3-year study on adolescent circadian development in wild-type and BLOC-1-deficient mice.
- Developed experimental methods, revealing 5-day faster sleep-wake consolidation in females vs males
- Benchmarked wheel-running against infrared (IR) sensor performance through time-series data analysis of behavioral activity tracking and circadian dynamics characterization.
- Authored and defended a 24-page thesis, earning UCLA Department Honors.

Silk Road Medical Inc. (Acquired by Boston Scientific) Minneapolis, MN
R&D Engineer Intern Jun 2021 - Sep 2021

- Engineered an Arduino-based carotid artery test bed and integrated sensors to assess catheter pressure and flow, enabling real-time clinician feedback during medical device evaluation.
- Automated FDA post-market device surveillance, accelerating root cause analysis and design iteration.

PROJECTS (kevinod.com/Projects)

- AI deep learning models: malaria cell object detection, breast cancer treatment survival modeling, and cardiovascular risk prediction with fairness calibration (>95% AUC-ROC) (2025).
- Pick-and-place 5-DOF robotic arm using ROS 2, MoveIt 2 path planning, and Gazebo sim (2025).
- Emergency endotracheal intubation model with Design Controls (Columbia Biomed Innovation 2022).
- Robotic pull-wire catheter planning & control algorithm for autonomous vascular navigation (2026).
- Full-stack, API-based medication reminder system for hospital EHR integration (UCLA YTP 2021).
- StarWalker Robot: CAD, 3D printing, hardware integration, Python, and PyBullet (Columbia 2023).

SKILLS

Engineering: Design Controls, Mechatronics, 3D Printing, Rapid Prototyping, Process Development

Software: Python, C++, MATLAB, Machine Learning, CAD (SolidWorks), Statistical Analysis, Excel